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Frequency
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Percentage



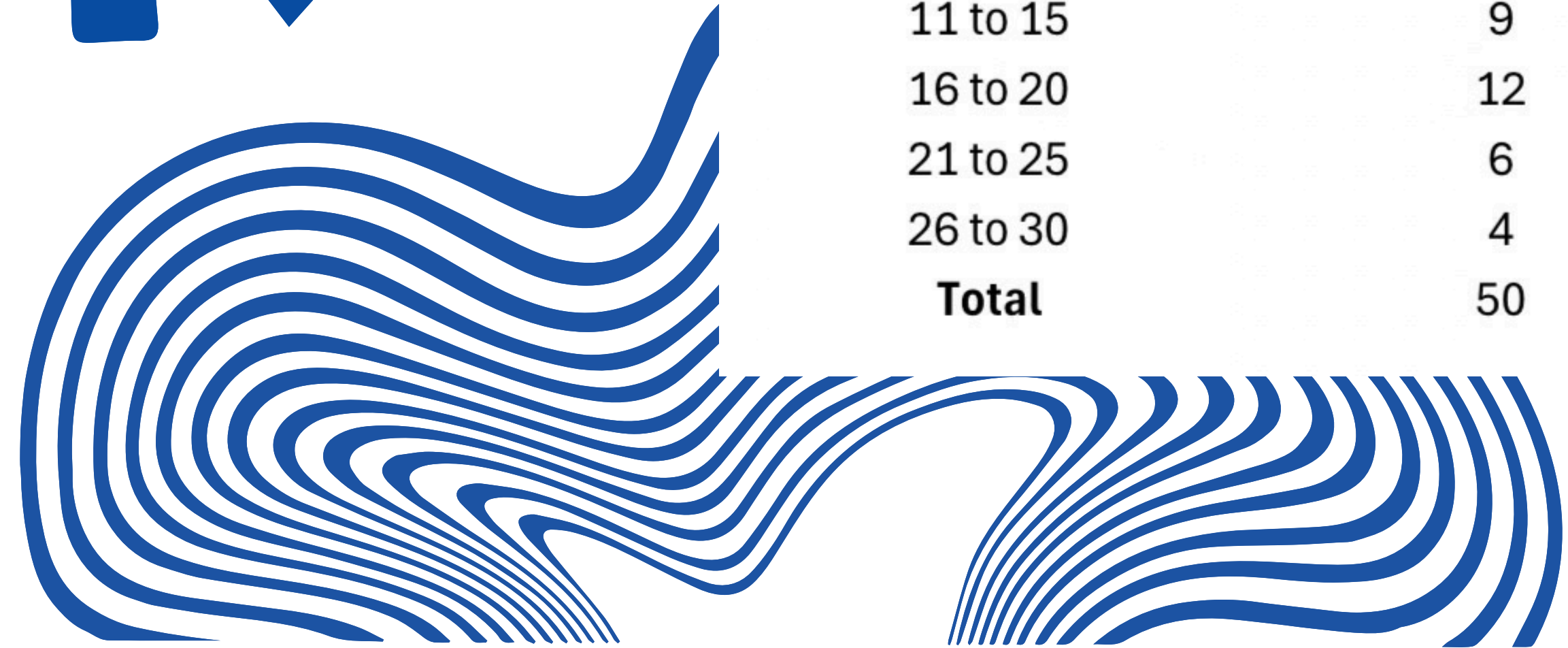
Distance of Tamaraws' Home to FEU Manila



A quantitative analysis of
students' distance from FEU
Manila using the principles of
normal distribution.

Presented By:

Cellona, Bryan Dion
Ricafort, Nathalie Faith



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Frequency Distribution

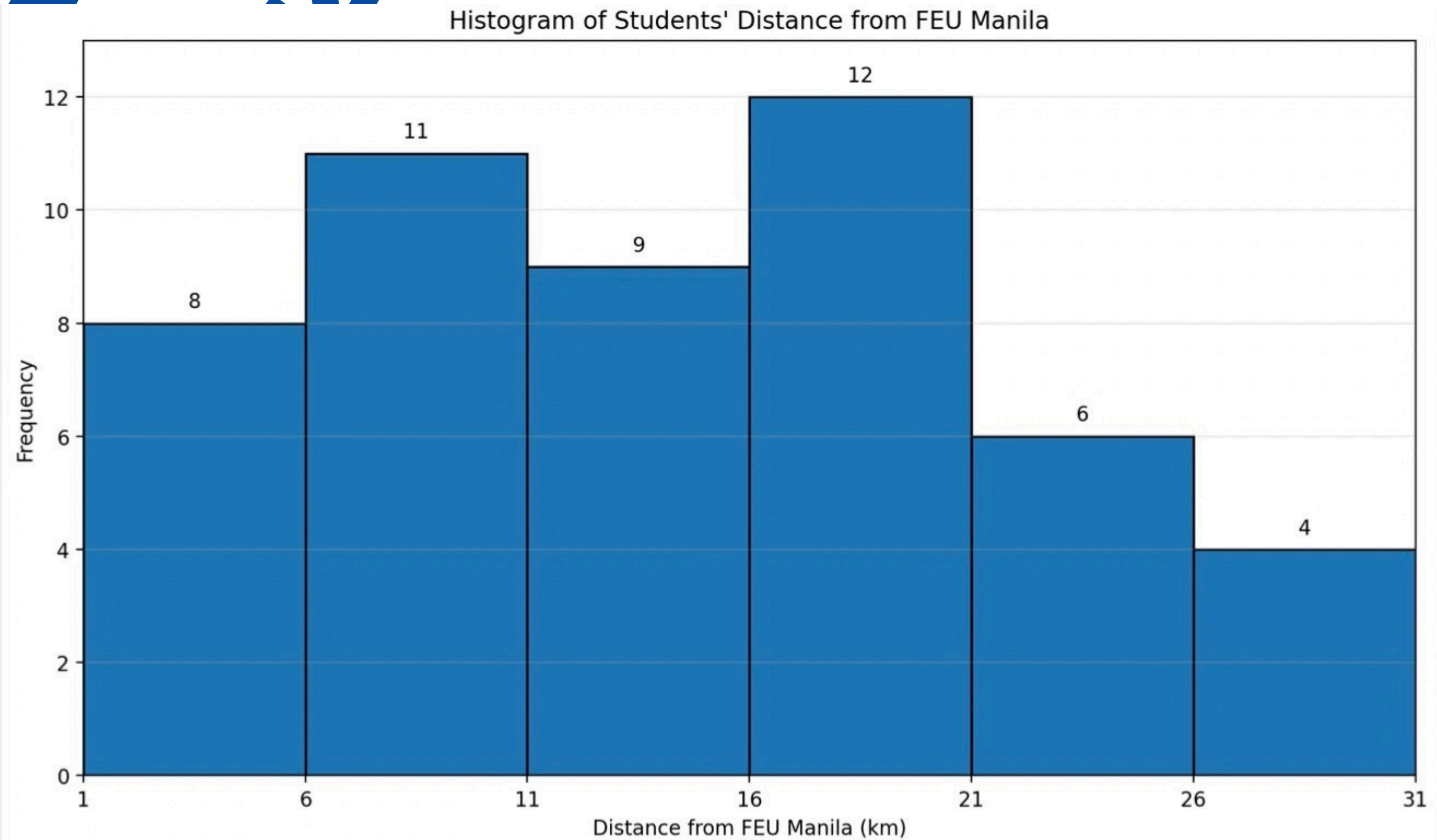
TABLE

Distance Group (km)	Frequency	Cumulative Frequency
1 to 5	8	8
6 to 10	11	19
11 to 15	9	28
16 to 20	12	40
21 to 25	6	46
26 to 30	4	50
Total	50	

Organized the 50 student responses into distance groups and recorded their frequencies.

Histogram

of Students' Distance from FEU Manila





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Mean & Standard Deviation

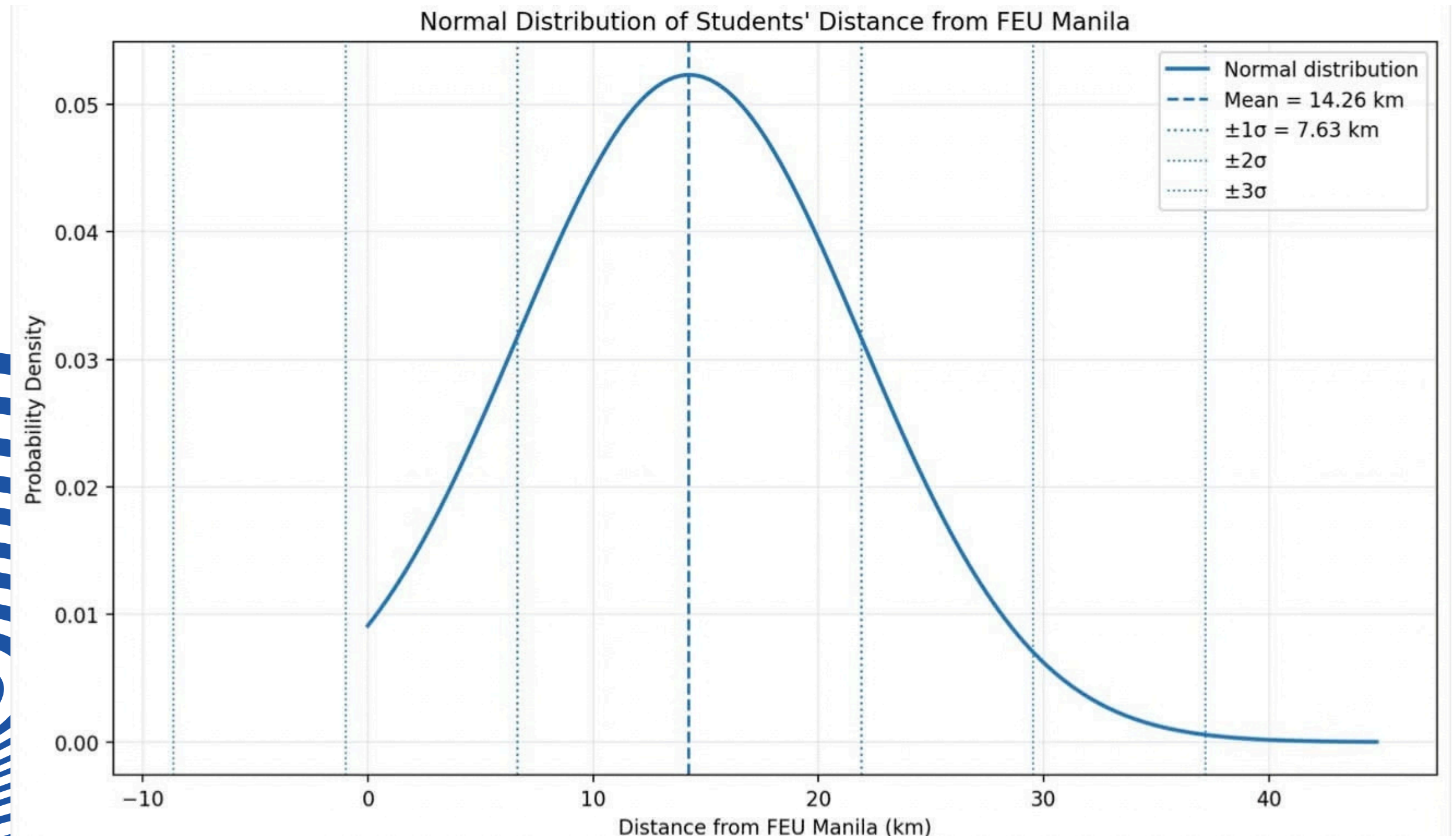
Mean (μ)	14.26
Standard Deviation (σ)	7.63

Calculated the average distance and the amount of variation in the data.



Created a normal distribution curve using the calculated mean and standard deviation to visualize the data's distribution.

Normal Distribution Curve



Percentage of the Data

Range	Calculation	Lower Limit (km)	Upper Limit (km)
Within 1σ	14.26 ± 7.63	6.63	21.89
Within 2σ	$14.26 \pm 2(7.63)$	-1	29.52
Within 3σ	$14.26 \pm 3(7.63)$	-8.63	37.15

Empirical Rule	Approx. Students (n=50)	Practical Range
68%	34	6.63–21.89 km
95%	48	0–29.52 km
99.7%	50	0–37.15 km

Applied the Empirical Rule: 68% → 95% → 99.7%



INTERPRETATION

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Is the distribution symmetric?

Not perfectly. The frequencies increase toward the middle groups and then decrease:

$8 \rightarrow 11 \rightarrow 9 \rightarrow 12 \rightarrow 6 \rightarrow 4$

This suggests that the data has a roughly bell-shaped tendency, but it is not perfectly symmetric like an ideal normal distribution.



Are there outliers?

There do not appear to be obvious extreme outliers. The observed distances range from 1 km to 30 km, and most observations are concentrated between the lower and middle distance groups.

What does the shape imply?

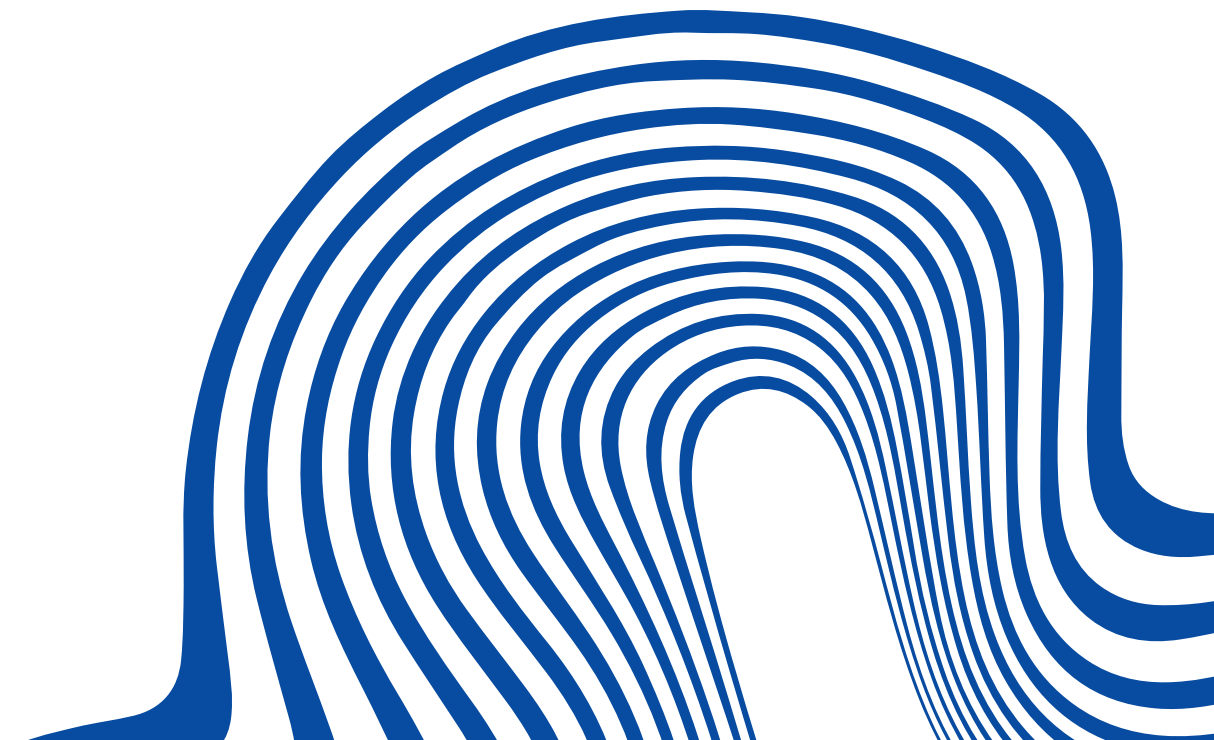
The distribution suggests that many students live relatively close to FEU Manila, while fewer students live at very large distances. The largest frequency is in the 16–20 km group, with 12 students.





How can this data be useful?

This information can help the school understand students' commuting situations. It could be useful when considering transportation options, class schedules, attendance concerns, and campus accessibility.





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